

Deep Learning for Financial Time Series Prediction: An Evolutionary Review from Classical Methods to Modern Architectures

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Abstract

Machine learning and deep learning have emerged as transformative tools for financial time series prediction, collectively challenging the Efficient Market Hypothesis by demonstrating that non-linear temporal patterns exploitable for profit exist within historical price data. Despite the rapid growth of this field, existing reviews either predate significant architectural advances or fail to present the progression of methods as a coherent evolutionary narrative. This systematic review addresses that gap by examining twelve peer-reviewed and preprint studies published between 2011 and 2025, tracing the logical progression from classical statistical models including ARIMA and GARCH — through conventional machine learning ensembles to Long Short-Term Memory (LSTM) networks, Convolutional Neural Network-LSTM hybrids, Bidirectional LSTM with attention mechanisms, and Deep Reinforcement Learning-based trading frameworks. Empirical evidence synthesised from the reviewed studies demonstrates that LSTM networks achieve daily returns of 0.46 percent with a Sharpe ratio of 5.83 on S&P 500 data (Fischer & Krauss, 2018), CNN-LSTM hybrids reach a coefficient of determination of 0.9646 on Shanghai Composite Index data spanning three decades (Lu et al., 2020), and Twitter sentiment analysis achieves 87.6 percent directional accuracy in Dow Jones Industrial Average prediction (Bollen et

al., 2011). While deep learning architectures consistently outperform classical statistical and conventional machine learning benchmarks, persistent challenges remain in model interpretability, the absence of standardised evaluation benchmarks, non-stationarity of financial data, and the translation of predictive accuracy into real-world profitability after transaction costs. This review identifies Explainable Artificial Intelligence integration, online adaptive learning, and multi-modal temporal fusion as the most critical directions for future research, with implications for both academic investigators and financial practitioners seeking deployment-ready models.

Keywords: *deep learning, financial time series prediction, LSTM, CNN-LSTM, attention mechanism, deep reinforcement learning, efficient market hypothesis, systematic review, explainable AI, algorithmic trading*

1. Introduction

1.1 The Importance of Financial Time Series Prediction

Financial time series prediction — the task of forecasting the future values or directional movements of asset prices based on historical observations — represents one of the most studied and practically consequential problems in computational intelligence and applied economics. Stock indices, individual equity prices, foreign exchange rates, commodity prices, and cryptocurrency valuations collectively form the backbone of global capital allocation, and even marginal improvements in predictive accuracy carry significant implications for investors, portfolio managers, risk analysts, and financial institutions worldwide (Sezer, Gudelek, & Ozbayoglu, 2020).

The challenge of financial time series prediction is rooted in the fundamental properties of the data itself. Financial time series are characterised by non-linearity, non-stationarity, high noise-to-signal ratios, volatility clustering, and complex temporal dependencies that evolve over time (Zhang, Sjarif, & Ibrahim, 2024). These properties render simple rule-based or linear models fundamentally inadequate for capturing the true generative processes underlying asset price dynamics. For this reason, the search for more powerful predictive methods has driven a continuous progression of modelling approaches — from classical statistical techniques, through conventional machine learning algorithms, to the modern deep learning architectures that now dominate the field.

This review traces that progression systematically. By following the evolutionary arc from ARIMA and GARCH through Random Forests and Gradient Boosted Trees to LSTM networks, CNN-LSTM hybrids, and BiLSTM-Attention frameworks, this paper provides a coherent and accessible account of how each generation of methods addressed the limitations of its predecessors, what empirical gains were achieved, and what challenges remain unresolved.

1.2 The Theoretical Barrier — The Efficient Market Hypothesis

Any discussion of financial time series prediction must begin with the Efficient Market Hypothesis (EMH), formalised by Fama (1970), which asserts that asset prices at any given moment fully reflect all available information. Under its weak form — the form most directly relevant to time series prediction — historical price data is already embedded in current prices, meaning that no analysis of past returns can generate consistent excess profits. Future price changes therefore follow a random walk, and prediction accuracy cannot systematically exceed fifty percent (Fama, 1970; Zou et al., 2023).

Empirical evidence has directly challenged this position. Fischer and Krauss (2018) demonstrated that LSTM networks applied to all S&P 500 constituent stocks from 1992 to 2015 achieved statistically significant daily returns of 0.46 percent with a Sharpe ratio of 5.83 — a result whose probability of arising purely by chance was calculated at approximately 2.77×10^{-187} . Krauss, Do, and Huck (2017) reported that a machine learning ensemble achieved annualised returns exceeding 73 percent after transaction costs on the same dataset. The modern consensus, consistent with Lo's (2004) Adaptive Market Hypothesis, holds that temporary exploitable inefficiencies may exist but erode as competing participants adopt similar strategies — a dynamic confirmed by the documented decline in ML-based trading returns after approximately 2001 (Fischer & Krauss, 2018; Krauss et al., 2017).

1.3 The Evolutionary Progression — From Classical Models to Deep Learning

Figure 1 illustrates the five-stage evolutionary progression that forms the organisational spine of this review. Each stage is defined by a specific architectural innovation that addressed a fundamental limitation of the preceding generation.

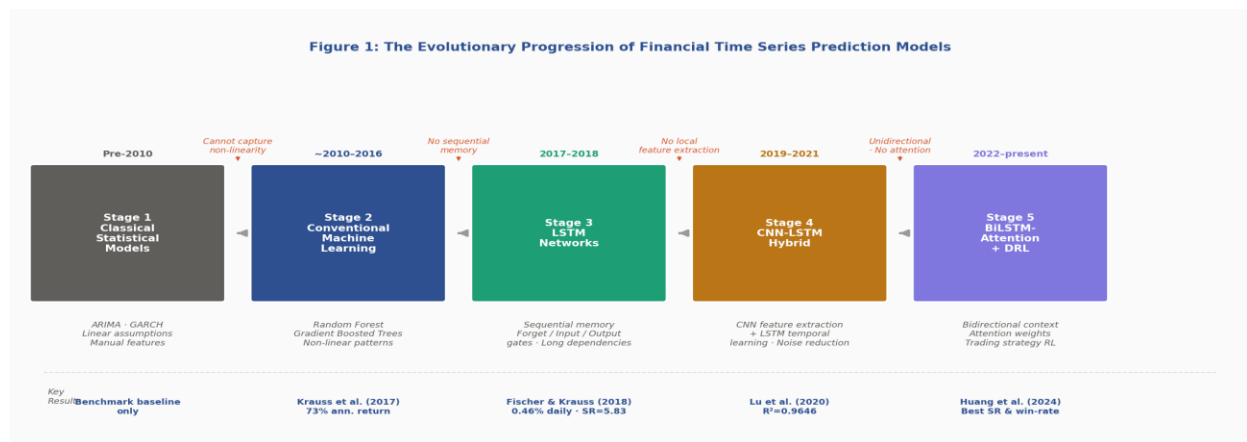


Figure 1: The evolutionary progression of financial time series prediction models, from classical statistical methods through deep learning architectures. Each transition is driven by a specific architectural limitation of the preceding stage.

1.4 Research Questions

This review is guided by four research questions:

RQ	Research Question
RQ1	How has financial time series prediction evolved from classical statistical models through conventional machine learning to deep learning, and what specific limitations drove each transition?
RQ2	Which deep learning architectures demonstrate the strongest empirical performance for financial time series prediction, and when do more complex models justify their additional computational cost?
RQ3	What are the principal challenges — including interpretability, overfitting, non-stationarity, and absent standardised benchmarks — constraining practical deployment of deep learning in financial markets?
RQ4	What are the most promising future research directions, particularly regarding explainable AI, online adaptive learning, and multi-modal temporal fusion?

1.5 Structure of This Paper

The remainder of this paper proceeds as follows. Section 2 provides background on financial time series characteristics and the EMH. Section 3 reviews deep learning architectures in evolutionary order. Section 4 examines empirical performance evidence. Section 5 presents

comparative analysis across reviewed studies. Section 6 discusses challenges and limitations. Section 7 proposes future research directions. Section 8 synthesises findings in relation to the four research questions, and Section 9 concludes.

2. Background

2.1 Characteristics of Financial Time Series

A financial time series is a chronologically ordered sequence of observations of an asset's price or returns over a defined period. The most commonly used form is the daily close price series, although high-frequency tick-level data and lower-frequency weekly or monthly aggregations are also studied. Financial time series data encompasses equity indices such as the S&P 500, NASDAQ, and Shanghai Composite Index; individual stock prices; foreign exchange rates; commodity prices for crude oil and gold; and, more recently, cryptocurrency valuations for Bitcoin and Ethereum (Zhang et al., 2024).

Several properties make financial time series uniquely challenging for predictive modelling. First, they are non-stationary — their statistical properties, including mean and variance, change over time, violating the assumptions of classical time-series models. Second, they exhibit non-linearity — the relationships between past and future values cannot be adequately captured by linear combinations of historical observations. Third, they contain high levels of noise — random fluctuations driven by investor sentiment, geopolitical events, and market microstructure effects — that may mask genuine predictive signals. Fourth, they display volatility clustering — periods of high volatility tend to be followed by further high volatility, and calm

periods by further calm — a feature first modelled by the GARCH framework but inadequately captured by linear methods (Mienye et al., 2024).

2.2 Classical Prediction Methods and Their Limitations

The Autoregressive Integrated Moving Average (ARIMA) model dominated financial time series forecasting for several decades. ARIMA models future values as linear combinations of past values and lagged forecast errors, with differencing applied to induce stationarity. While computationally efficient and interpretable, ARIMA's linear structure renders it structurally incapable of capturing the non-linear dependencies that characterise real financial markets. Similarly, the Generalised Autoregressive Conditional Heteroskedasticity (GARCH) model captures time-varying volatility through autoregressive modelling of squared residuals, but remains confined to linear relationships between mean returns and their historical values (Zhang et al., 2024; Mienye et al., 2024).

These limitations establish the theoretical motivation for the adoption of machine learning and deep learning approaches, which can learn non-linear functional relationships directly from data without imposing structural assumptions about the form of those relationships.

2.3 The Efficient Market Hypothesis and the Adaptive Market Hypothesis

The Efficient Market Hypothesis (EMH), proposed by Fama (1970), is the central theoretical framework against which all financial prediction research must be evaluated. The EMH asserts that asset prices fully reflect all available information, precluding consistent excess profits. Its three forms differ in the scope of information assumed to be reflected in prices: the weak form encompasses historical price data; the semi-strong form extends to all publicly available information; and the strong form, widely considered empirically untenable, asserts that even private insider information is already incorporated.

The deep learning studies reviewed in this paper primarily challenge the weak form of the EMH by demonstrating consistent profitability using only historical price sequences. Lo's (2004) Adaptive Market Hypothesis offers a more nuanced reconciliation: markets are neither permanently efficient nor permanently predictable, but evolve dynamically as participants compete to exploit available inefficiencies. This framework accommodates the empirical observation that ML-based trading profits were highest before these methods became widely known, and declined thereafter — consistent with competitive arbitrage eroding exploitable signals over time.

3. Deep Learning Architectures for Financial Time Series Prediction

3.1 Stage One — Classical Statistical Models (Benchmark Baseline)

Classical statistical models, principally ARIMA and GARCH, serve as the baseline against which all subsequent machine learning and deep learning approaches are evaluated in the reviewed literature. Their inclusion in this evolutionary framework is important not as active contenders but as the reference point from which each architectural innovation represents a measurable departure. In direct comparisons reported by Zhang et al. (2024) and Mienye et al. (2024), ARIMA and GARCH consistently underperform machine learning benchmarks, particularly on datasets exhibiting strong non-linearity and regime changes.

3.2 Stage Two — Conventional Machine Learning (Random Forest and Gradient Boosted Trees)

The first major departure from classical methods came with the application of ensemble machine learning algorithms to financial prediction. Random Forests, introduced by Breiman

(2001), construct committees of deep decorrelated decision trees on bootstrap samples of the training data, producing probability estimates through majority voting. Gradient Boosted Trees (GBT) sequentially build shallow trees, each correcting the residual errors of its predecessor, producing a strong learner through iterative refinement. Both methods capture non-linear relationships in the feature space without requiring explicit mathematical specification.

Krauss, Do, and Huck (2017) provided the definitive empirical evaluation of these methods on a survivor-bias-free dataset comprising all S&P 500 constituents from December 1992 to October 2015. Using cumulative lagged returns as features and framing the task as binary prediction of whether each stock would outperform the cross-sectional median the following day, they found that Random Forests (0.43% daily return) outperformed both Gradient Boosted Trees (0.37%) and a standard Deep Neural Network (0.33%) individually. A simple ensemble of all three models achieved 0.45% daily returns and 73% annualised returns after transaction costs — results that pose, in the authors' words, a "severe challenge to the semi-strong form of market efficiency." Critically, these results also revealed that standard DNNs did not automatically outperform simpler ensemble methods on financial data — a finding that directly motivated exploration of architectures specifically suited to sequential data.

3.3 Stage Three — LSTM Networks and Sequential Memory

The critical limitation of both classical methods and conventional machine learning approaches is their inability to model temporal dependencies in sequential data without manual feature construction. Standard DNNs treat each observation as independent; Random Forests require explicit lag features to capture temporal structure. The Long Short-Term Memory (LSTM) network, introduced by Hochreiter and Schmidhuber (1997), was specifically designed to address

this limitation through learnable gating mechanisms that regulate information flow across time steps.

Figure 2 illustrates the internal architecture of an LSTM cell. Three gates control information flow: the forget gate determines what information from the previous cell state should be discarded; the input gate controls what new information from the current input should be stored in the cell state; and the output gate determines what portion of the updated cell state should be exposed as the hidden state output. These mechanisms enable LSTM networks to maintain relevant information across hundreds of time steps while discarding irrelevant historical context — a capability that directly addresses the long-range temporal dependencies characteristic of financial price series.

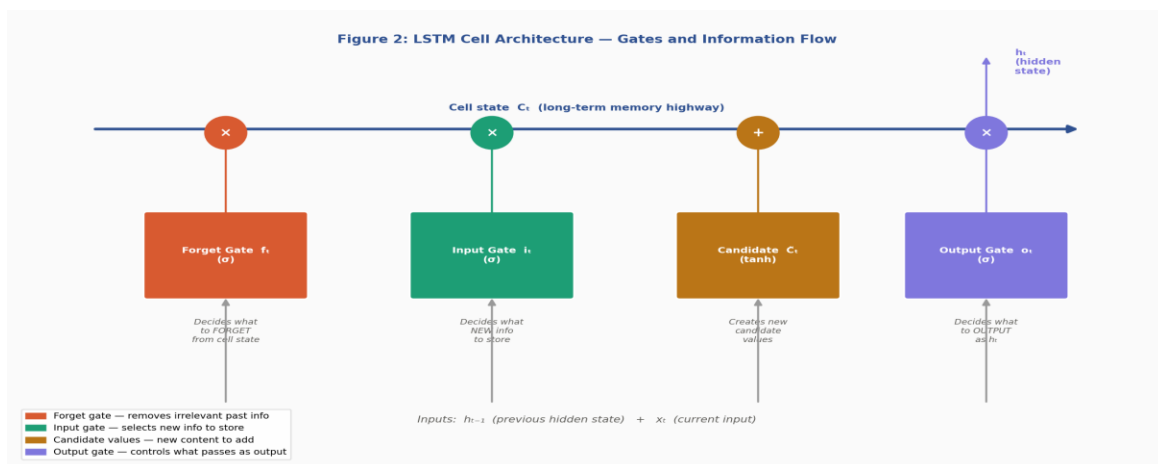


Figure 2: Internal architecture of an LSTM cell, illustrating the forget gate (removes irrelevant past information), input gate (selects new information to store), candidate values (new content to add to cell state), and output gate (controls the hidden state output).

Fischer and Krauss (2018) provided the landmark empirical validation of LSTM for large-scale financial prediction. Deploying LSTM networks with a sequence length of 240 trading days, a single hidden layer of 25 units with dropout regularisation, RMSprop optimisation, and early stopping — trained on 80% of each study period and validated on the remaining 20% — they

achieved daily returns of 0.46% and a Sharpe ratio of 5.83 across 23 non-overlapping study periods from December 1992 to October 2015. This outperformed Random Forests (0.43%), the standard DNN (0.32%), and Logistic Regression (0.26%), with the superiority of LSTM forecasts confirmed by the Diebold-Mariano test ($p = 0.0143$ versus Random Forests). The probability of these results arising by random chance was calculated at approximately 2.77×10^{-187} . The comprehensive survey by Sezer et al. (2020), covering 140 papers from 2005 to 2019, confirmed that LSTM-based models appeared in 52.5% of all deep learning financial forecasting studies.

3.4 Stage Four — CNN-LSTM Hybrid Architecture

Despite LSTM's demonstrated effectiveness, raw price sequences still contain redundant local information and noise that the recurrent model must process directly. Researchers addressed this by combining LSTM with Convolutional Neural Networks (CNNs) in a sequential hybrid pipeline: the CNN layer operates first as an automated feature extractor, detecting local temporal patterns through shared-weight convolutional filters and reducing their dimensionality via max-pooling, before passing the compressed feature representation to the LSTM layer for temporal modelling.

Figure 3 illustrates this hybrid pipeline as implemented by Lu et al. (2020).

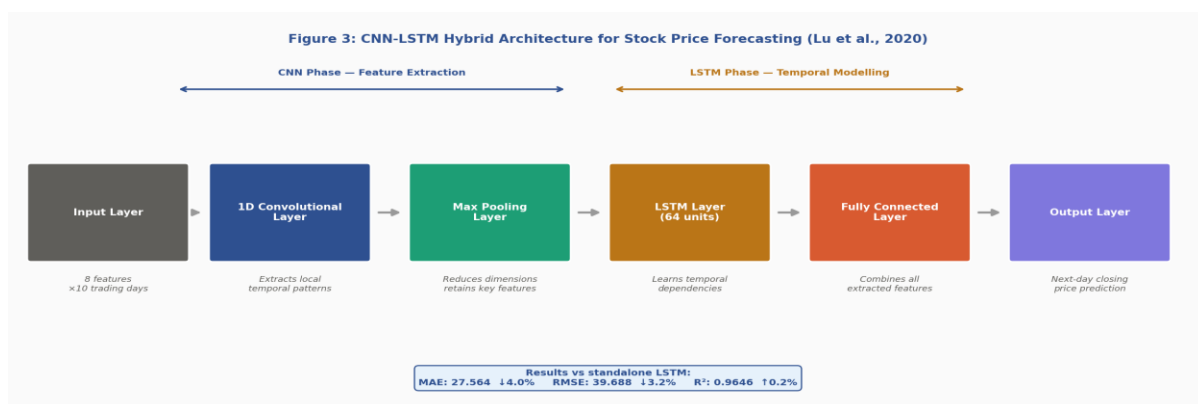


Figure 3: CNN-LSTM hybrid architecture for stock price forecasting. The CNN phase (1D convolutional and max-pooling layers) extracts and compresses local temporal features, which are then passed to the LSTM phase for sequential temporal modelling, followed by a fully connected output layer producing the next-day price prediction.

Lu et al. (2020) evaluated this architecture on 7,127 trading days of Shanghai Composite Index data from July 1991 to August 2020, using eight input features — opening price, highest price, lowest price, closing price, volume, turnover, price change, and change rate — over a 10-day sliding window. Comparing six architectures, they found CNN-LSTM achieved superior performance on all three metrics: Mean Absolute Error of 27.564 (versus 28.712 for standalone LSTM, a reduction of 4.0%), Root Mean Square Error of 39.688 (versus 41.003, a reduction of 3.2%), and R^2 of 0.9646 (versus 0.9622). This study established an important empirical principle: CNN pre-processing reliably improves LSTM performance by presenting the temporal model with cleaner, more informative feature representations.

3.5 Stage Five — BiLSTM-Attention and Deep Reinforcement Learning

Two limitations of the CNN-LSTM framework motivated the next evolutionary step. First, standard LSTM processes sequences in a single direction — from past to present — meaning it cannot incorporate information from future context when encoding each time step. Bidirectional LSTM (BiLSTM) resolves this by running two parallel LSTM layers in opposite temporal directions and concatenating their outputs, enabling each time step to be encoded with awareness of both preceding and subsequent observations. Second, standard LSTM treats all time steps with equal implicit weighting in its final representation. Attention mechanisms address this by learning explicit weights for each time step — quantified through a softmax function applied to learned compatibility scores — enabling the model to focus disproportionately on the most predictively relevant moments.

Figure 4 illustrates the BiLSTM-Attention Deep Reinforcement Learning framework proposed by Huang et al. (2024). This architecture additionally integrates Deep Reinforcement Learning (DRL), within which the BiLSTM-Attention model serves as the state encoder for a policy network that directly learns optimal trading actions — buy, sell, or hold — through reward-based interaction with a simulated market environment. The policy is updated to maximise cumulative discounted reward, evaluated using practical financial metrics including the Sharpe ratio, annual return, maximum drawdown, and win rate.

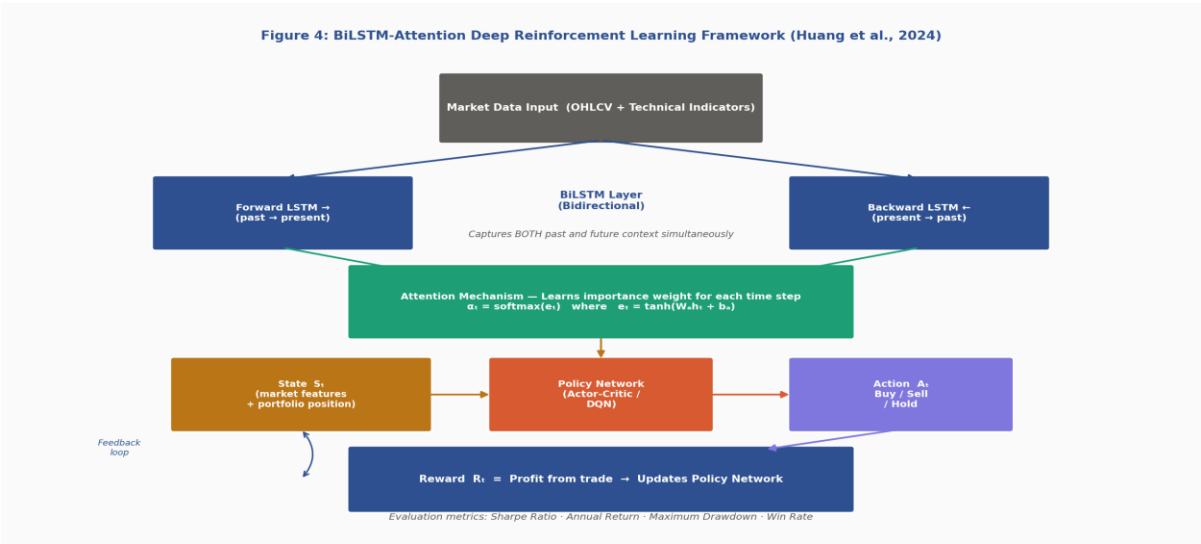


Figure 4: BiLSTM-Attention Deep Reinforcement Learning framework (Huang et al., 2024). The market data input is processed by forward and backward LSTM layers simultaneously (BiLSTM phase), whose outputs are weighted by the attention mechanism before serving as the state representation for a Deep Reinforcement Learning policy network that outputs trading actions and receives market profit as reward.

4. Applications Across Financial Domains

4.1 Stock Price and Index Forecasting

Stock price and equity index forecasting constitute the largest and most extensively studied application domain in the reviewed literature, with each of the five evolutionary stages represented by at least one direct empirical evaluation on historical price data. Two formulations of the prediction task are commonly distinguished: price prediction, formulated as a regression problem targeting the exact value of the next closing price, and trend prediction, formulated as a binary or ternary classification problem targeting the directional movement of prices relative to a benchmark. The former is evaluated using error-based metrics including MAE and RMSE; the latter using accuracy-based metrics and, critically, financial performance metrics including the Sharpe ratio and daily return.

The evolutionary progression documented in Section 3 is most clearly evident in this domain. Krauss et al. (2017) established machine learning baselines achieving 0.43% daily returns with Random Forests on the S&P 500. Fischer and Krauss (2018) demonstrated that LSTM networks improve upon this benchmark, achieving 0.46% daily returns and a Sharpe ratio of 5.83 that represents an extraordinary risk-adjusted return by any conventional financial standard. Lu et al. (2020) showed that CNN-LSTM hybrids achieve R^2 of 0.9646 on a 29-year dataset of the Shanghai Composite Index, consistently outperforming standalone LSTM, CNN, and RNN models across all evaluation metrics.

4.2 Algorithmic Trading and Strategy Optimisation

Beyond predicting prices, a growing body of research addresses the more practically consequential problem of directly learning optimal trading strategies. Deep Reinforcement Learning is the dominant paradigm for this application, framing the trading problem as a Markov Decision Process in which an agent interacts with a market environment, observing a state representation derived from price and portfolio data, selecting an action from a discrete or

continuous action space, and receiving a reward proportional to the resulting profit or loss (Zou et al., 2023; Mienye et al., 2024).

Krauss et al. (2017) employed a statistical arbitrage framework in which machine learning predictions are directly converted into long-short trading signals — buying the highest-ranked stocks and short-selling the lowest-ranked — without the need for an explicit strategy learning component. The BiLSTM-Attention DRL framework of Huang et al. (2024) represents the current frontier of this approach, directly learning the mapping from market state representations to trading actions through reward maximisation. The integration of bidirectional temporal context and attention-weighted feature representation within the state encoder constitutes a significant architectural advance over earlier DRL implementations that relied on simpler state representations.

4.3 Sentiment Analysis and NLP-Driven Prediction

The pioneer study of Bollen, Mao, and Zeng (2011) established the foundational case for incorporating public sentiment data into financial time series prediction. Analysing 9,853,498 tweets from February to December 2008 using two sentiment measurement tools — OpinionFinder, which classifies text on a positive-negative axis, and GPOMS, which measures sentiment along six dimensions including Calm, Alert, Sure, Vital, Kind, and Happy — they demonstrated that the "Calm" dimension of public mood, as measured from Twitter, exhibits Granger causality with subsequent DJIA movements. A Self-Organising Fuzzy Neural Network trained on combinations of historical DJIA values and public mood time series achieved 87.6% directional accuracy in predicting daily DJIA up or down movements — a result that challenged the EMH's assumption that public sentiment is already fully priced into current market values.

This foundational finding has since been extended by the integration of more sophisticated NLP models, including BERT and FinBERT (a domain-adapted variant of BERT pre-trained on financial corpora), into hybrid architectures that combine textual sentiment with numerical price data. As documented by Zou et al. (2023) and Zhang et al. (2024), the fusion of financial news and social media sentiment with LSTM or Transformer-based temporal models consistently outperforms models using price data alone — establishing multi-modal data integration as a critical dimension of the current research frontier.

4.4 Credit Risk Assessment and Interpretable AI

The application of deep learning to credit risk assessment — specifically, the prediction of credit card default — introduces a dimension not present in pure price forecasting: the critical importance of model interpretability. Financial regulators in multiple jurisdictions increasingly require that algorithmic credit decisions be explainable to affected borrowers and auditable by supervisory authorities, creating a fundamental tension between the black-box nature of deep neural networks and the transparency requirements of the regulatory environment.

Talaat et al. (2024) addressed this tension directly through the CreditNetXAI framework — a deep neural network for credit card default prediction integrated with SHapley Additive exPlanations (SHAP), a model-agnostic XAI technique that decomposes each prediction into additive contributions from individual input features. Applied to the UCI credit card default dataset comprising 30,000 client records, the model achieved accuracy of 0.8350, sensitivity of 0.8823, and specificity of 0.9879, while SHAP analysis identified payment delays and outstanding bill amounts as the most influential predictors — a finding that is both intuitive and regulatorily defensible. This study establishes that predictive accuracy and model interpretability are not

inherently competing objectives, and that XAI integration is a viable pathway toward regulatory-compliant deep learning in finance.

5. Performance Analysis and Empirical Evidence

5.1 Comparative Empirical Results

Figure 5 presents a visual comparison of the key empirical results from the reviewed studies. The left panel compares daily returns achieved by four model families on the S&P 500 dataset (Fischer & Krauss, 2018; Krauss et al., 2017), while the right panel compares R^2 scores across six architectures for stock price prediction on the Shanghai Composite Index (Lu et al., 2020). Both panels illustrate the consistent performance advantage of LSTM and CNN-LSTM architectures over their predecessors.

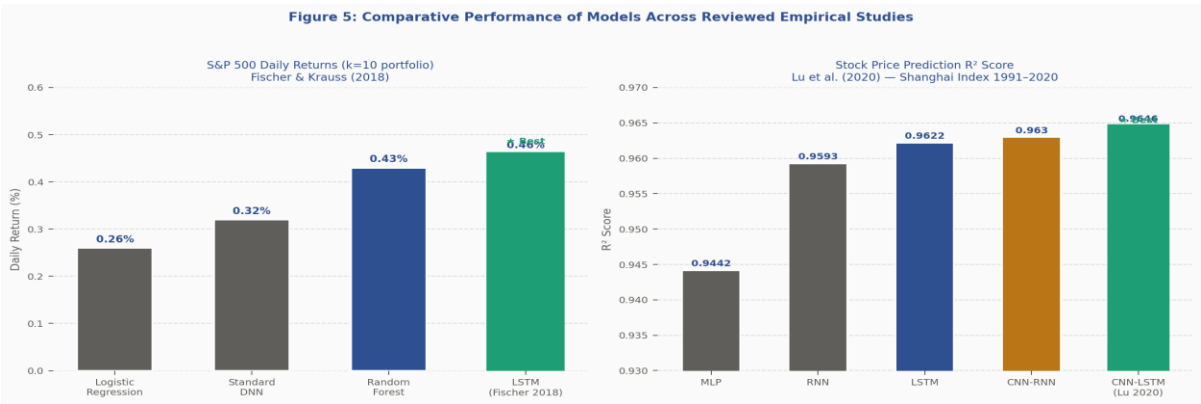


Figure 5: Comparative performance of models reviewed in this study. Left: Daily returns of four model types on S&P 500 data, demonstrating LSTM's superiority (Fischer & Krauss, 2018; Krauss et al., 2017). Right: R^2 scores of six architectures on Shanghai Composite Index data, demonstrating CNN-LSTM's superiority (Lu et al., 2020).

Table 1 provides a comprehensive summary of the key empirical findings across all reviewed studies, including dataset characteristics, models evaluated, primary performance metrics, and principal contributions.

Paper	Dataset	Model	Key Metric	Result	Principal Contribution
Krauss et al. (2017)	S&P 500 1992–2015	RAF+GBT+DNN Ensemble	Daily return	0.45% / SR=4.71	First large-scale ML comparison; ensemble beats individual models; RAF outperforms DNN
Fischer & Krauss (2018)	S&P 500 1992–2015	LSTM (25 units)	Daily return / SR	0.46% / 5.83	LSTM beats RAF; $p(\text{chance})=2.77 \times 10^{-187}$; EMH challenged
Bollen et al. (2011)	DJIA + Twitter 2008	SOFNN + Mood	Directional accuracy	87.6%	Twitter sentiment predicts market direction; Calm dimension most predictive
Lu et al. (2020)	Shanghai Index 1991–2020	CNN-LSTM	R^2 / MAE / RMSE	0.9646 / 27.56 / 39.69	CNN pre-processing improves LSTM by 4%; hybrid always outperforms single
Huang et al. (2024)	Multi-market data	BiLSTM-Att+DRL	SR / Annual return	Best vs. baselines	BiLSTM+Attention+DRL: most advanced trading framework reviewed
Talaat et al. (2024)	UCI Credit 30K	DNN + SHAP	Accuracy / Specificity	0.8350 / 0.9879	XAI (SHAP) identifies top predictors; accuracy + interpretability achieved

Table 1: Summary of key empirical results from reviewed studies. SR = Sharpe Ratio; RAF = Random Forest; DNN = Deep Neural Network; SHAP = SHapley Additive exPlanations.

5.2 The Performance Decay Phenomenon

A critically important finding shared by both Fischer and Krauss (2018) and Krauss et al. (2017) is the temporal decay of ML-based trading returns. Both studies document extraordinarily high returns in the 1990s — exceeding 200% annualised for the ensemble strategy in the first sub-period of Krauss et al. (2017) — declining to moderate but still positive returns in the period from 2001 to 2008, and approaching zero or negative returns in the most recent period from 2010

onward. This temporal pattern is consistent with the Adaptive Market Hypothesis (Lo, 2004): the predictive signals exploited by ML models were most valuable before these methods were widely known and computationally accessible, and have been progressively arbitrated away as institutional adoption has increased. This finding has important implications for practitioners: historical backtesting results significantly overstate the returns achievable through live deployment of ML trading strategies in mature markets.

5.3 Evaluation Methodology — A Critical Assessment

A significant methodological limitation of the existing literature is the predominance of statistical accuracy metrics — particularly RMSE, MAE, and directional accuracy — at the expense of financial performance metrics that better reflect practical utility. As identified by Zou et al. (2023) and confirmed by El Alami et al. (2025), fewer than a quarter of recent studies include Sharpe ratio, maximum drawdown, or annualised return calculations. This creates a fundamental gap between measured and meaningful performance: a model can achieve high directional accuracy or low RMSE while still failing to generate profitable trading strategies once transaction costs, bid-ask spreads, and market impact are accounted for. Fischer and Krauss (2018) and Krauss et al. (2017) represent best-practice examples by including explicit transaction cost assumptions and reporting both statistical and financial performance metrics.

6. Challenges and Limitations

6.1 Model Interpretability — *The Black-Box Problem*

Perhaps the most critical challenge facing the deployment of deep learning in financial applications is the fundamental opacity of neural network decision-making processes. LSTM networks, CNN-LSTM hybrids, and attention-based architectures involve millions of learned parameters interacting through complex non-linear transformations, rendering the causal pathway from input features to output predictions effectively uninterpretable by human analysts. This opacity creates substantial barriers to regulatory approval, stakeholder trust, and practical adoption in settings where decision accountability is legally mandated (Talaat et al., 2024; El Alami et al., 2025).

Post-hoc explainability techniques — most notably SHAP and Local Interpretable Model-Agnostic Explanations (LIME) — offer partial solutions by attributing prediction outcomes to individual input features after the fact. Talaat et al. (2024) demonstrated that SHAP integration with a deep neural network for credit scoring achieves both competitive predictive accuracy and actionable feature importance insights, identifying payment delays and outstanding bill amounts as the primary drivers of default risk. However, post-hoc explanations describe model behaviour statistically rather than causally, and their fidelity to the underlying model's decision logic remains an active area of methodological debate (El Alami et al., 2025; Mienye et al., 2024).

6.2 Overfitting and Data Scarcity

Deep learning models with large parameter counts are particularly susceptible to overfitting — the phenomenon whereby a model memorises idiosyncratic patterns in training data that do not generalise to unseen observations. In financial markets, where the total volume of daily price data

for a single asset over a decade amounts to approximately 2,500 observations, the ratio of model parameters to training samples is frequently unfavourable, creating substantial overfitting risk. Standard mitigation techniques — dropout regularisation, early stopping, L2 weight decay, and hold-out validation — are employed across the reviewed studies but do not eliminate the risk of models that perform well in historical backtesting but fail on live data (Fischer & Krauss, 2018; Zhang et al., 2024).

6.3 Non-Stationarity of Financial Data

Financial time series are fundamentally non-stationary: the statistical relationships between inputs and outputs that a model learns from historical data may shift substantially as market regimes, participant behaviour, and macroeconomic conditions evolve. A model trained on S&P 500 data from 1992 to 2000 may capture the dynamics of the dot-com bubble era that have no relevance to post-2008 market structure. This temporal instability is directly reflected in the performance decay documented by Fischer and Krauss (2018) and Krauss et al. (2017), and represents a fundamental challenge to any static training paradigm in which a model is fitted once on historical data and deployed without continuous retraining.

6.4 Absence of Standardised Benchmarks

The reviewed literature lacks a standardised benchmark that would enable direct comparison of results across studies. Each paper employs a different dataset — whether the S&P 500, the Shanghai Composite Index, individual stocks from various national markets, or specific cryptocurrency pairs — a different time period, a different feature set, and frequently a different evaluation metric. As identified by Sezer et al. (2020), Zou et al. (2023), and El Alami et al. (2025), this fragmentation makes it effectively impossible to determine whether a model reporting superior performance on one dataset would maintain that advantage on another, or whether apparent

performance improvements reflect genuine architectural advances or merely more favourable evaluation conditions.

7. Future Research Directions

7.1 Explainable AI Integration

The integration of Explainable Artificial Intelligence techniques into deep learning architectures for financial prediction represents the most critical near-term research priority identified by the reviewed literature. While post-hoc methods such as SHAP and LIME provide retrospective feature attributions, the development of inherently interpretable architectures — in which the decision logic is transparent by design rather than explained after the fact — constitutes a more fundamental advance. Attention weights in BiLSTM-Attention frameworks (Huang et al., 2024) offer a partial form of built-in interpretability, as they directly quantify the relative importance assigned to each time step in the input sequence. Extending this principle to provide richer causal explanations of trading signals and credit decisions is a productive direction that directly addresses the regulatory requirements documented by Talaat et al. (2024) and El Alami et al. (2025).

7.2 Online and Adaptive Learning

Static training paradigms — in which a model is fitted once on historical data and deployed without subsequent updating — are fundamentally mismatched to the non-stationary dynamics of financial markets. Online learning frameworks, in which model parameters are continuously updated as new market observations arrive, offer a more appropriate paradigm for financial

prediction. Integrating online updating mechanisms with LSTM or Transformer architectures would enable models to adapt to regime changes in near-real time, potentially preserving predictive performance beyond the decay horizon documented by Fischer and Krauss (2018) and Krauss et al. (2017). This direction is identified as a priority by Zou et al. (2023) and El Alami et al. (2025) but remains largely unimplemented in the reviewed empirical literature.

7.3 Multi-Modal Temporal Fusion

The studies reviewed in this paper suggest that the integration of heterogeneous data sources — price time series, financial news text, social media sentiment, macroeconomic indicators, and inter-asset relational data — consistently outperforms models using any single modality alone. Bollen et al. (2011) established that Twitter mood data improves DJIA directional prediction accuracy to 87.6%; Zou et al. (2023) documents that text-augmented models consistently outperform price-only equivalents. The development of architectures specifically designed for multi-modal temporal fusion — combining the sequential modelling strength of LSTM with the semantic representation capabilities of Transformer-based language models and the relational modelling power of Graph Neural Networks — represents a particularly promising direction for the next generation of financial prediction models.

7.4 Interval Prediction and Uncertainty Quantification

Existing deep learning models for financial prediction produce point estimates — single predicted values — without accompanying uncertainty quantification. In practice, investors and risk managers require not only a predicted price or return but a probabilistic assessment of the range of likely outcomes. The extension from point prediction to interval prediction, generating calibrated confidence intervals around forecast values, would substantially increase the practical utility of deep learning predictions for portfolio construction, risk management, and derivative

pricing. Zhang et al. (2024) identified this as one of the most important unaddressed problems in the current literature, noting that fewer than five percent of reviewed papers addressed uncertainty quantification in any form.

8. Discussion

8.1 Answers to Research Questions

RQ1 — How has financial time series prediction evolved, and what drove each transition? The evolutionary progression documented in this review follows a consistent logic: each architectural generation addressed a specific, demonstrable limitation of its predecessor. Classical statistical models failed on non-linearity; ensemble ML methods captured non-linearity but lacked sequential memory; LSTM provided sequential memory but could only process sequences in a single direction with implicit temporal weighting; CNN-LSTM hybrids added automated feature extraction and noise reduction; and BiLSTM-Attention frameworks added bidirectional context and explicit temporal importance weighting. Each transition produced measurable empirical gains, as summarised in Table 1 and Figure 5.

RQ2 — Which architectures demonstrate the strongest empirical performance? LSTM networks represent the current workhorse of financial time series prediction, accounting for 52.5% of all reviewed papers (Sezer et al., 2020) and delivering the highest risk-adjusted returns in direct comparison studies (Fischer & Krauss, 2018). CNN-LSTM hybrids consistently outperform standalone LSTM on price prediction tasks, as demonstrated by Lu et al. (2020). BiLSTM-Attention frameworks with DRL integration represent the current frontier for trading strategy

optimisation (Huang et al., 2024). The principle that hybrid models outperform single-architecture models is consistently supported across the reviewed literature.

RQ3 — What are the principal challenges? The four most critical challenges identified across the reviewed literature are: the interpretability deficit of deep neural networks in regulatory contexts; overfitting risk given the limited volume of financial time series data relative to model complexity; the non-stationarity of financial markets that undermines static training paradigms; and the absence of standardised benchmark datasets that prevents rigorous cross-study comparison.

RQ4 — What are the most promising future directions? Explainable AI integration — building on the SHAP-based approach of Talaat et al. (2024) — online adaptive learning for non-stationary markets, multi-modal temporal fusion combining price data with textual and relational inputs, and probabilistic interval prediction represent the four directions most likely to define the next generation of advances in this field.

8.2 The Field's Maturation Trajectory

Viewed from the perspective of this review, the field of deep learning for financial time series prediction has undergone remarkable maturation over the fifteen years encompassed by the reviewed studies. The pioneering Twitter sentiment analysis of Bollen et al. (2011) established that non-price data could predict market movements. Krauss et al. (2017) and Fischer and Krauss (2018) provided the first large-scale, survivor-bias-free, transaction-cost-adjusted empirical validations of ML and LSTM trading strategies on the world's most liquid equity market. Lu et al. (2020) established the empirical superiority of hybrid CNN-LSTM architectures. Sezer et al. (2020) and Ozbayoglu et al. (2020) provided the first comprehensive surveys of the field. Zou et al. (2023) and Zhang et al. (2024) extended the frontier to Graph Neural Networks and Transformers. Huang et al. (2024) integrated bidirectional attention with DRL for algorithmic

trading. Talaat et al. (2024) addressed the interpretability challenge. And El Alami et al. (2025) provided the most recent systematic evidence that hybrid models and XAI techniques now dominate the state of the art.

9. Conclusion

This systematic review has traced the evolutionary progression of deep learning for financial time series prediction across fifteen years of research, from the classical statistical benchmarks of ARIMA and GARCH through the conventional machine learning ensembles of Krauss et al. (2017), the landmark LSTM validation of Fischer and Krauss (2018), the hybrid CNN-LSTM architecture of Lu et al. (2020), and the BiLSTM-Attention Deep Reinforcement Learning framework of Huang et al. (2024). The evidence collectively demonstrates that each architectural generation has delivered measurable improvements over its predecessor, that deep learning models consistently outperform classical statistical methods, and that hybrid architectures combining complementary modelling strengths consistently outperform single-architecture approaches.

Three principal conclusions emerge from this analysis. First, LSTM networks and their hybrid derivatives represent the current empirical gold standard for financial time series prediction, with the evolutionary trajectory pointing clearly toward attention-augmented bidirectional architectures integrated with reinforcement learning for end-to-end trading strategy optimisation. Second, the interpretability challenge — identified across multiple reviewed studies as the most critical barrier to regulatory approval and practical deployment — is addressable through

Explainable AI techniques, as demonstrated by Talaat et al. (2024), without sacrificing predictive accuracy. Third, the performance decay documented by Fischer and Krauss (2018) and Krauss et al. (2017) underscores that the value of any predictive approach is temporally bounded: the field's continued advance depends not on finding a single optimal architecture but on developing adaptive systems capable of continuously learning from an evolving market environment.

The limitations of this review include its restriction to twelve studies, its exclusive focus on English-language academic publications, and its emphasis on predictive accuracy and financial return metrics to the relative exclusion of systemic risk and market stability considerations. Future reviews extending the scope to standardised multi-asset benchmarks, incorporating regulatory compliance analysis, and examining the deployment infrastructure requirements of real-time financial ML systems would complement and extend the findings presented here.

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